Microplastic Concentration Mapping in Freshwater

Two Practical Measurement Tracks: Live Boat Monitoring and Portable Shoreline Post-processing

Year 3 Group Project – Working background and concept-selection review

October 2026

Abstract

This revised background review combines the previously screened freshwater literature, a quantitative two-hour 400-m^2 mission screen, and a 49-method-family re-screen with the stakeholder's requirement to produce a portable, automated, two-dimensional map of microplastic (MP) concentration. Two workflows are evaluated: (A) real-time measurements from a boat and (B) geotagged water collection with rapid, self-contained processing at the shoreline. **Routine access to FTIR, Raman, or a chemistry laboratory is not assumed.** Camera-assisted particle counting, flow imaging, electrical impedance, fluorescence, polarisation, and water-quality proxies are compared by operating principle, deployment effort, specificity and evidence. A staged decision plan is proposed to distinguish *suspected particle concentration* from a defensible *microplastic concentration* estimate. Paper counts in this review are preliminary keyword-screening outcomes rather than demonstrated method performance.

1 Stakeholder requirements and scope

The stakeholder brief calls for a system that estimates microplastic concentration over a lake or river surface, produces a *two-dimensional spatial concentration map*, surveys relatively large areas quickly, is practical to deploy and can be validated against appropriate reference measurements. The briefing explicitly allows direct measurement or image-based methods and a boat or aerial platform. It notes that conventional collection, filtration and microscope counting are slow and have limited spatial coverage. Here we take the boat as the vehicle and reformulate measurement into two feasible deployment tracks. **A continuous coloured image of water appearance is not, by itself, a validated map of microplastic concentration.** Similarly, GPS only places estimates on a map; it does not establish that they represent plastic particles [1], [2], [3].

The current engineering constraint is that **routine specialist laboratory access is impractical**. This makes a compact, cleanable, low-maintenance measurement path and a robust portable cross-check especially important. Low-cost field processing can support preliminary concentration estimates, while polymer-specific identification would still need occasional external testing if definitive MP attribution is required.

1.1 What is meant by concentration?

For counted particles, the target quantity is $C = N_{\mathrm{MP}}/V_{\mathrm{water}}$ in particles/L, stated for a defined detectable particle-size range and sampling depth. A count of *suspected* MPs inferred from images should be reported as such until specificity has been demonstrated. For model-derived estimates, uncertainty and calibration domain must be reported. Polymer mass concentrations from thermal analysis are a different measurand and cannot simply be substituted for particles/L [3], [4].

2 Literature selection and four background themes

From the prior Web of Science–Scopus search, 18,942 records were identified. After removing 7,318 duplicates and 727 document-type exclusions, 10,897 remained for title/abstract screening; 10,833 were preliminary include candidates and 64 needed manual review. The four theme tags overlap: causes (6,430), water effects (7,575), laboratory/measurement methods (6,654) and data trends/mapping (5,320). These labels do not represent completed full-text eligibility screening.

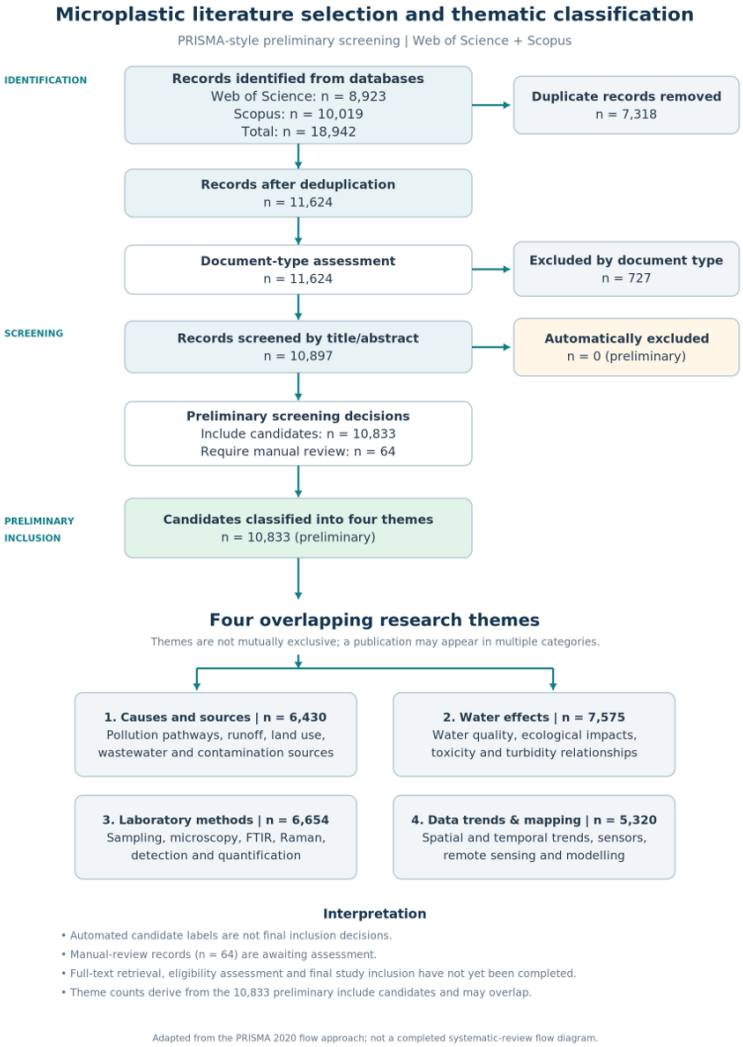


Figure 1: Original preliminary screening and four overlapping classifications. Counts are not independent study totals.

2.1 Sources and causes of riverborne microplastics

Land use, wastewater, roads, runoff and local discharge influence the spatial distribution of MPs. Tropical river studies illustrate variation between sites and sampling periods rather than one uniform concentration across a catchment [1], [5], [6], [7]. For either track, each measurement must retain sampling location, date, depth and nearby environmental context.

2.2 Effects and physical/chemical water properties

Relevant literature covers ecological effects and associations with turbidity, conductivity, suspended solids and nutrients. These properties may influence sensor signals but are *not uniquely attributable to MPs*. For instance, nitrogen/conductivity relationships in surface-water research do not demonstrate a universal electrical MP concentration sensor [8], [9]. Therefore sediment, algae, bubbles, dissolved salts and changing illumination must be tested explicitly as interferents.

2.3 Measurement and reference methods

Collection, filtration, microscope imaging, FTIR and Raman feature prominently in the initial screening. Image-based counting can quantify visible particles after appropriate segmentation, whereas spectroscopy can support polymer identity but requires equipment beyond the present project constraint [10], [11], [12], [13]. Portable microscope imaging is therefore a feasible preliminary *field-reference* for particle counts, but not a definitive chemical reference for all MPs.

2.4 Data trends and spatial prediction

Rivers change with rainfall, flow and location, so isolated point measurements must be mapped with times and uncertainty. Remote imagery and GIS have been explored, but translating visible surface patterns or turbidity into MP particles/L requires paired calibration and external testing [1], [9], [14]. *Track B can create a post-survey heat map from discrete geotagged samples; a heat map does not require that concentrations were measured while the boat was moving.*

3 Two-track system architecture

3.1 Two processes, one map specification

Figure 2 presents two end-to-end workflows. The diagram is intentionally technology-agnostic: its purpose is to define the operational stages and thereby the criteria used to filter candidate sensing methodologies. The boat and GPS path are shared, but the sensing stage and the downstream handling occur in different places. Track A is more automated and ambitious; Track B is a practical alternative when continuous in-water particle detection is unreliable, too expensive or difficult to maintain.

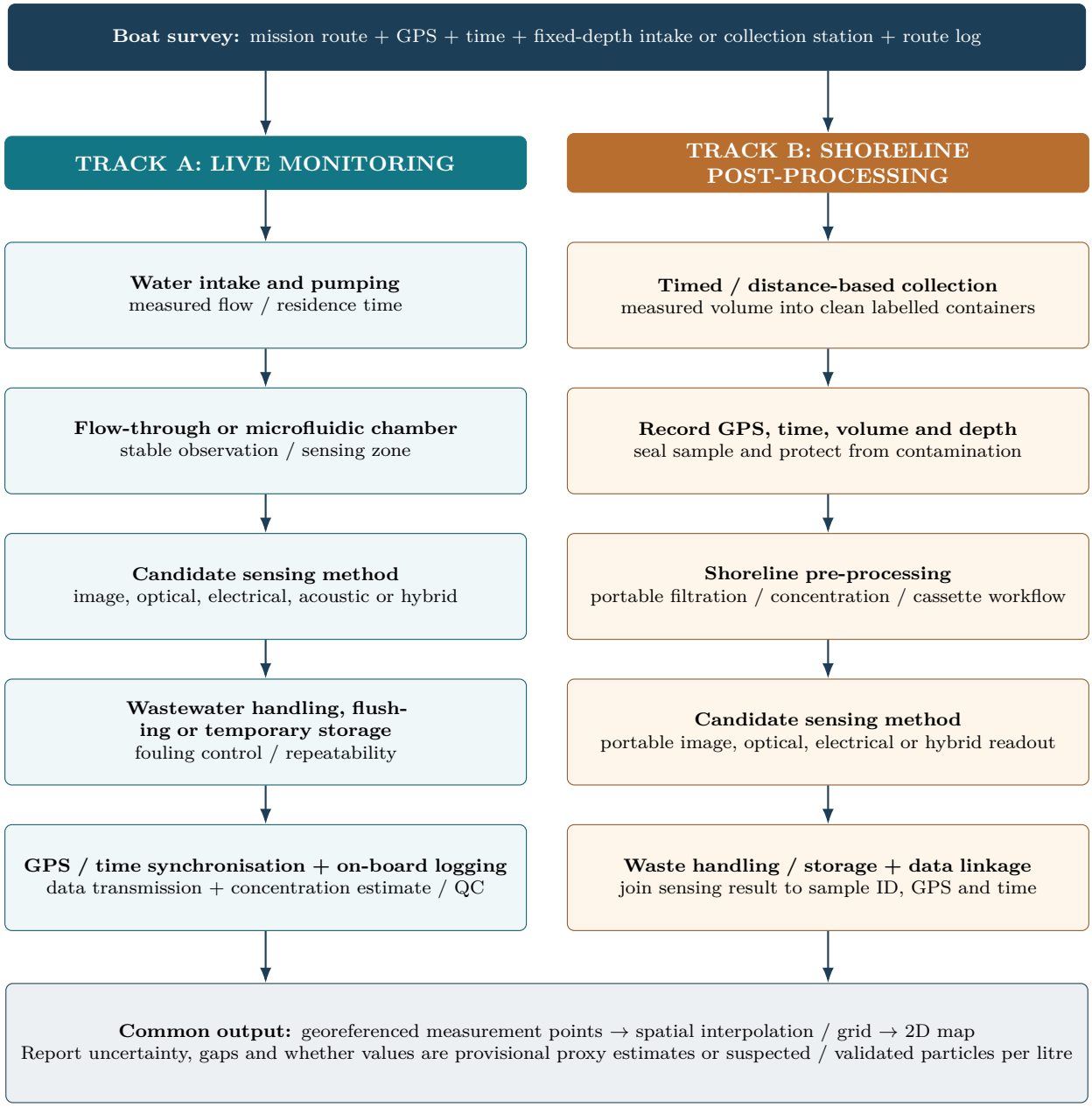


Figure 2: Two deployable tracks that meet the briefing's mapping goal without assuming a specific sensing technology. The "candidate sensing method" box is the screening stage at which methods are filtered against portability, automation, contamination control, throughput, interpretability and map-readiness. Track B is batch/near-real-time after retrieval, not continuous in-situ sensing.

The flowchart is therefore a *method-selection framework*. A sensing method should only be retained if it fits the required stage in the workflow and satisfies the briefing's constraints: portability, sufficiently rapid surveying, a defensible relationship to actual MP concentration, manageable contamination control, and practical linkage to GPS-referenced map generation [15]. In the sections below, the literature is reviewed using these process requirements rather than assuming from the outset that any single technique (for example camera + AI or EIS) must be adopted.

3.2 Track A: live monitoring from the boat

A pump draws a measured stream through a cleanable observation channel. A fixed-focus camera and controlled LED illumination acquire frames at a known effective observed volume; segmentation and tracking distinguish successive particles from repeat appearances. GPS position should be assigned to the *time the water entered the intake*, correcting for tubing/flow delay. The output can be updated at short intervals while the boat moves. Camera systems are supported by microscope-image AI and micro-flow imaging literature, though performance in untreated, moving, turbid river water is not guaranteed [11], [12], [16].

Optional adjunct: multi-frequency impedance or polarised scattering may add non-image features, but neither resolves chemical identity automatically. Bulk water conductivity should not be labelled "MP concentration" without an independently validated model [3], [17], [18], [19].

3.3 Track B: automatic collection and portable shoreline processing

At predetermined locations or distances the boat collects a *known* volume in a labelled, low-contamination sample container. Back at the shoreline, each sample is filtered on a portable membrane or processed in a sealed cassette, then imaged with a USB/digital microscope. An AI counter produces a count, and an operator reviews borderline particles and blanks. Pre-treatment or Nile Red staining is an optional method-development branch, not a necessary routine step: staining, rinsing, imaging filters and waste management must be considered [11], [20], [21], [22]. This route decouples sampling from difficult live optics and permits re-counting archived images.

3.4 Common mapping software and quality requirements

Both tracks store the same minimum fields: sample ID, latitude/longitude, intake time, actual water volume, sampling depth, particle-size cutoff, device model/version, quality flags and concentration estimate. A point/segment map should be shown before interpolation. If interpolation is used, choose a declared method (e.g., inverse-distance weighting or kriging if data support it), retain actual sampled points and assess its error against withheld points. A smooth heat map does not imply unsampled regions have been measured.

4 Literature review organised around the process and screening evidence

4.1 What methods have actually been investigated?

The broad Laboratory Methods keyword screen comprises 6,654 candidate records; its 14 labels describe *mentions* rather than verified instrument deployment. Figure 3 shows the complete distribution. Here, those methods are interpreted against the workflow stages in Fig. 2: intake and chamber compatibility, sensing principle, waste/storage needs, data-linkage demands and readiness for rapid mapping. The chart therefore helps narrow which methods deserve further consideration, rather than prescribing a single sensor in advance.

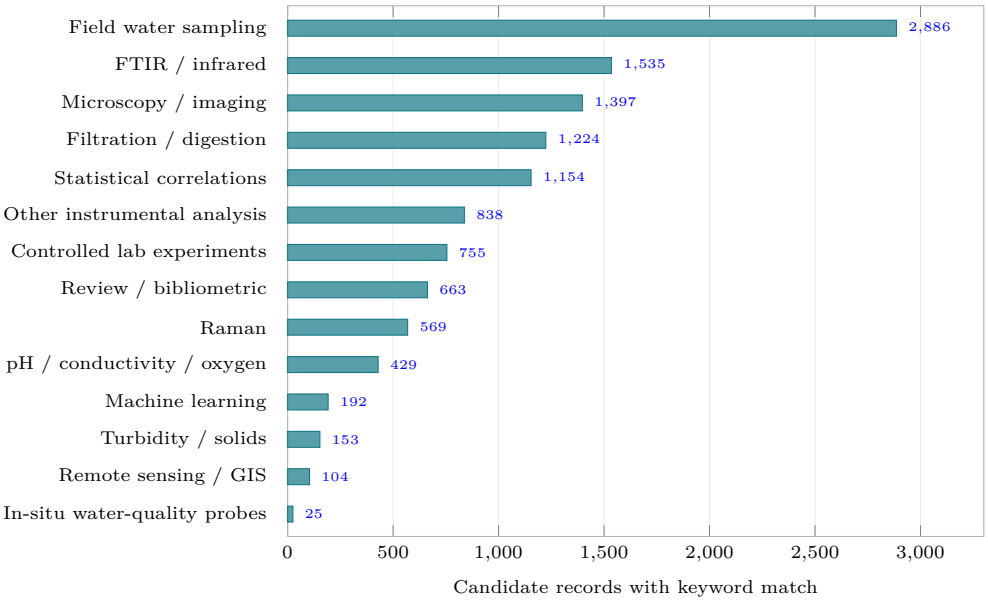


Figure 3: Original 14-category method screen ($n = 6{,}654$): overlapping title/abstract matches, not confirmed experimental application or accuracy.

4.2 Targeted sensing and processing technologies

A complementary search across all 10,833 candidates (title/abstract/keyword fields) found relatively few mentions of impedance/EIS (5), microfluidics (22), holography (13) and polarimetric scattering (3), compared with electrochemical sensing (55), Nile Red (138) and FTIR (1,909). Ultrasound/sonication (42) includes sample preparation;

these 42 are *not* 42 ultrasonic MP counters. Figure 4 keeps this screen separate from the broader 6,654-record method classification.

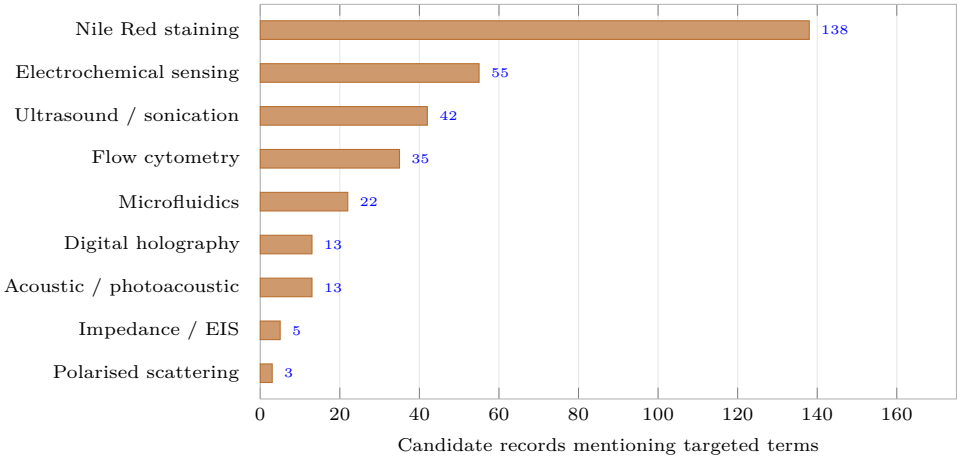


Figure 4: Targeted method vocabulary in 10,833 preliminary candidates; overlap and non-detection uses are possible.

5 Reproducible identification, filtering and elimination of sensing methodologies

5.1 Scope and definition of an eligible sensor

For reproducibility, we re-ran a new, purpose-built, multi-label search on the **10,833 original preliminary include candidates**, scanning title, abstract, author keywords and Keywords Plus (where present). The companion Python script records each regular-expression rule, its frequency, and the titles/DOIs of all matching publications. The inventory contains **49 explicitly defined method terms** in five families; no method is assumed to be experimentally deployed simply because the term occurs. These revised rules are narrower than some earlier broad-screen rules; hence the new frequencies are **not directly interchangeable** with the original 14-category counts, even when labels sound similar.

The second distinction is functional rather than bibliometric: *collection/preparation*, *direct or particle-related readout*, *specialist reference analysis*, *non-specific water-quality proxy*, and *data processing/mapping* play different roles. A filter, a particle concentrator and a GIS algorithm may be essential components of a useful system, but **none is a stand-alone MP concentration sensor**. Laboratory reference methods can remain informative literature even if specialist instruments cannot be included as routine field equipment [3], [4], [10].

5.2 Screening funnel and disposition accounting

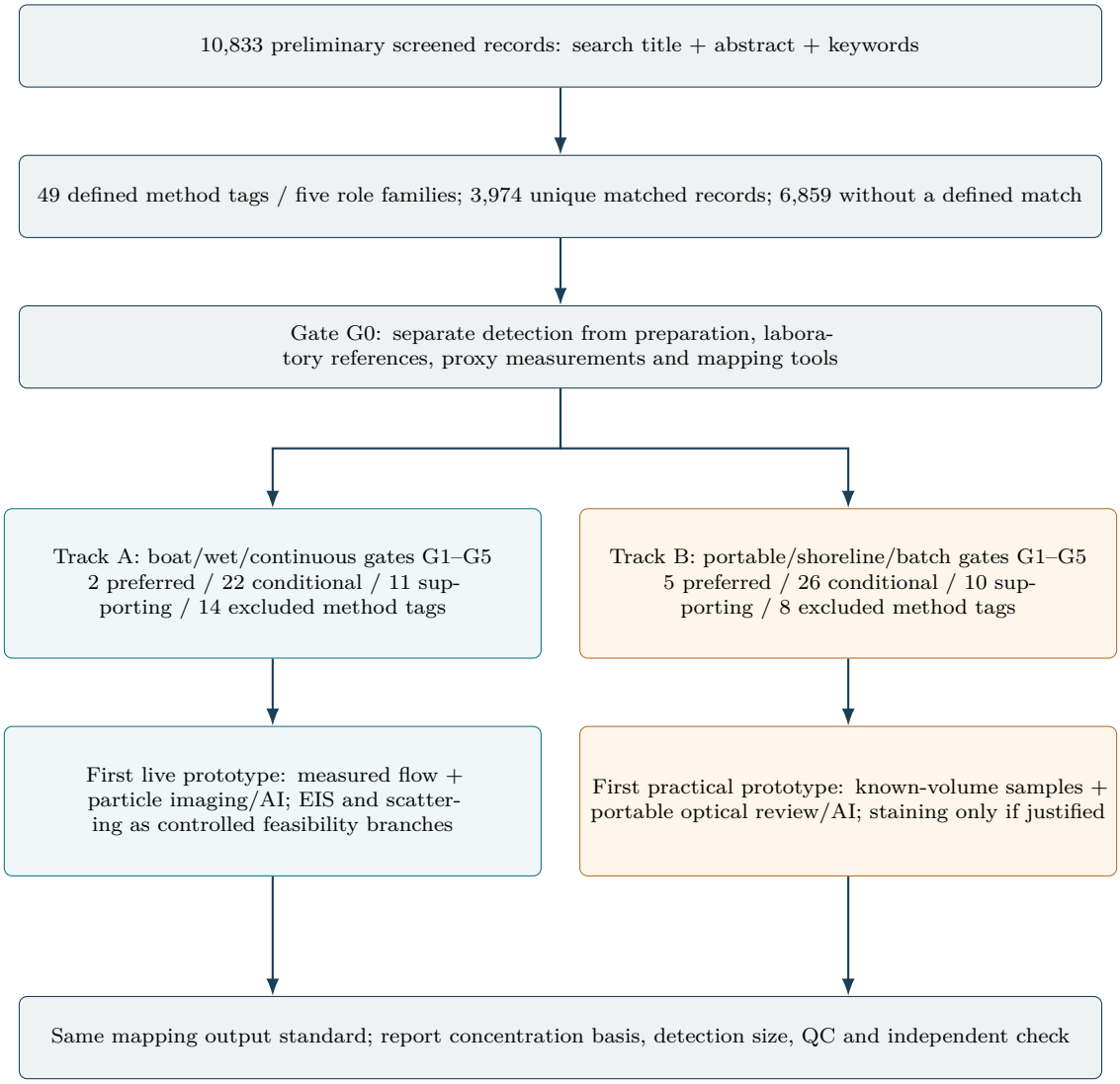


Figure 5: Reproducible method term screen followed by engineering gates; disposition counts refer to *method tags*, not distinct papers or fully validated sensing devices. Unmatched records remain accessible for further screening.

5.3 Sequential gates, rather than popularity-based selection

Table 1: Method-elimination stages derived from the stakeholder specification. The criteria are testable design gates, not published accuracy thresholds.

Gate	Main question	Track A: live boat	Track B: shoreline post-processing
G0	Is the technique a sensing output?	Differentiate detector from pump, filter, separator or prediction software.	Same distinction; standalone reference tools are allowed as components.
G1	Can it yield a concentration estimate?	Measured observed volume or validated calibration from readout to particles/L, with a declared size cutoff.	Count linked to collected and analysed fraction; account for blank and recovery.
G2	Can it fit the process?	Wet continuous/near-continuous operation in a cleanable flow path; GPS/intake-delay linkage.	Portable batch processing, manageable turnaround, traceable sample IDs and GPS.
G3	Can it withstand real water?	Test salts, sediment, algae, bubbles, light/temperature changes, fouling.	Test background, processing loss and between-sample contamination.
G4	Is field use feasible?	Safe power, robust components, automation and manageable waste.	No routine specialist laboratory; affordable and safe shoreline equipment/waste handling.
G5	Can its result be audited?	Raw optical/electrical signals retained; uncertainty and QC flags.	Archive original samples/images where practical, blind recounts, blanks and confidence levels.

Disposition rules. **P** denotes preferred primary route; **C**, technically plausible but awaiting proof; **S**, supporting component/covariate; and **X**, excluded as a routine core instrument under the present no-specialist-lab constraint.

"X" is *not* a claim that the method has low scientific validity, and "P" is *not* a claim of demonstrated microplastic selectivity. A method can be retained as a complement but eliminated as the *only* sensor for particles/L.

5.4 Complete defined method-family census

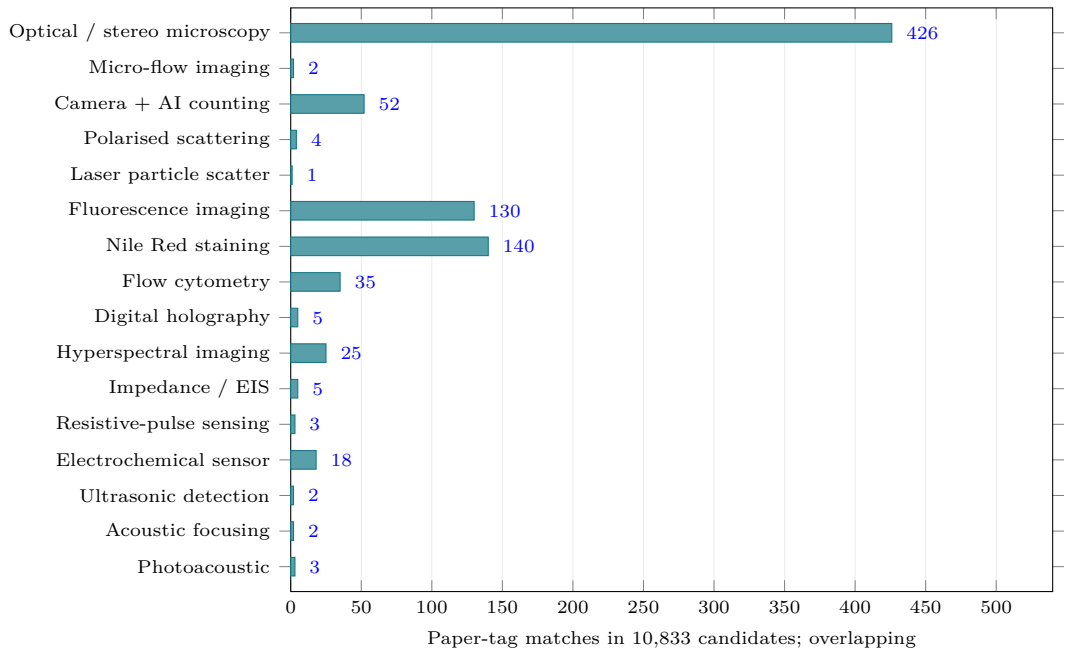


Figure 6: Potential optical, electrical and acoustic particle-readout families; ultrasound sample preparation is intentionally excluded from the direct-detector chart.

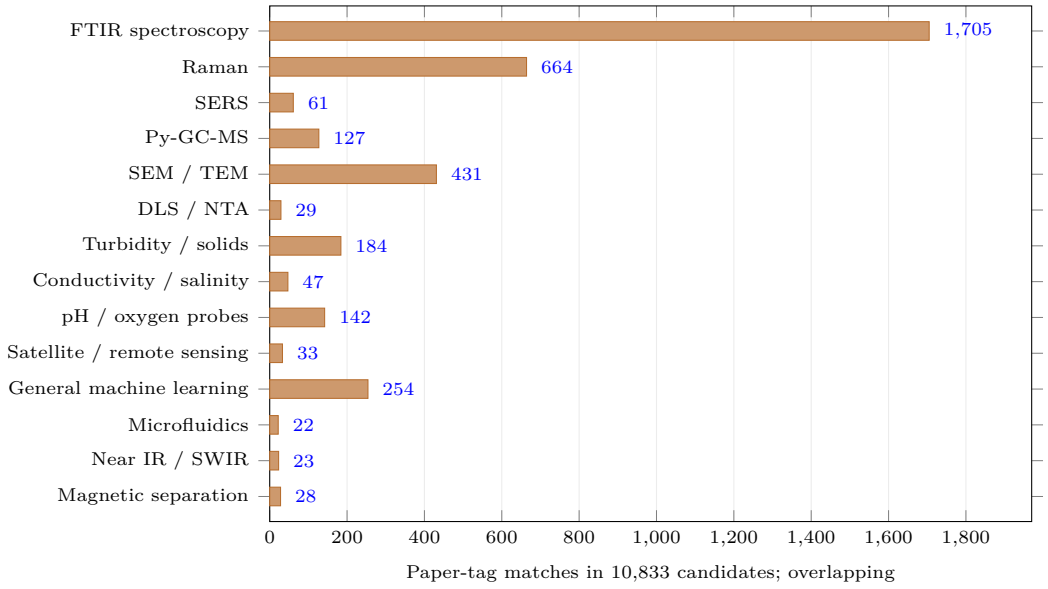


Figure 7: Reference techniques, pretreatment, ancillary probes and model/mapping tools: prevalence of terms does not imply field sensing capability.

The first figure compares particle-readout technologies; the second shows reference, auxiliary and analysis terms. Together with the older broad 14-category figure, these distinguish *literature prominence* from *operational selection*. For example, FTIR and Raman tags are common because the techniques are useful for polymer identification, but routine bench spectroscopy fails the portable, rapid and low-infrastructure gate. Conversely, few impedance or flow-imaging mentions do not imply that they are unworkable; instead they signal a smaller and less mature direct evidence base [3], [11], [17], [18].

Table 2: Expanded method identification across the original 10,833 papers. Matches are title/abstract/keyword terms, may overlap, and are not proof that the authors applied the method.

ID	Explicitly screened method family	Papers	A	B	Constraint / decision rationale
Collection / pretreatment					
M01	Manual field grab / surface sampling	1,122	S	P	volume/contamination control
M02	Pump / automated water intake	13	S	S	flow calibration & carry-over
M03	Membrane / vacuum filtration	107	C	P	time & membrane recovery
M04	Density / flotation separation	373	X	C	reagents and waste
M05	Chemical / enzymatic digestion	113	X	C	chemical safety/time
M06	Ultrasonic sample sonication	10	X	C	preparation, not detection
M41	Microfluidics / lab-on-chip handling	22	S	S	microchannels need antifouling/volume control
M46	Magnetic / electrostatic separation	28	X	C	functionalisation / recovery controls
Optical particle readout					
M07	Brightfield / stereo optical microscopy	426	C	P	particle appearance not polymer proof
M08	Flow-through / micro-flow particle imaging	2	P	C	volume/focus & duplicates
M09	Camera + AI counting / segmentation	52	P	P	MP-like images vs true MPs
M10	Polarised light scattering / polarimetry	4	C	C	sediment/algae discrimination
M11	Laser / nephelometric particle scattering	1	C	C	optical non-specificity
M12	Fluorescence / laser induced fluorescence	130	C	C	photobleaching/matrix
M13	Nile Red fluorescent staining	140	X	C	wet chemistry, false positives
M14	Flow cytometry / fluorescence cytometry	35	C	C	expensive/complex; identity uncertain
M15	Digital holography / inline holography	5	C	C	optical alignment, inference
M16	Hyperspectral / multispectral laboratory imaging	25	C	C	cost/illumination/background
M17	UV-visible absorbance / colorimetry	3	S	S	matrix dependence
M47	Smartphone / handheld digital microscopy	0	C	P	low cost; resolution and repeatability
M49	Optical coherence tomography (OCT)	1	C	C	cost/penetration/selectivity
Electrical / acoustic					
M18	Electrical impedance spectroscopy (EIS)	5	C	C	salt/electrode artefacts
M19	Resistive pulse / Coulter / microchannel sensing	3	C	C	throughput/clogs, size cut-off
M20	Electrochemical / voltammetric sensing	18	C	C	electrode specificity & fouling
M21	Dielectrophoresis / capacitive particle sensing	73	C	C	field geometry and salinity
M22	Microwave / RF dielectric resonance	1	C	C	water permittivity dominates
M23	Ultrasonic attenuation / scattering detection	2	C	C	particle/background contrast
M24	Acoustic focusing / acoustophoresis	2	C	C	enrichment only, not counting
M25	Photoacoustic sensing / spectroscopy	3	C	C	specialised transducers/lasers
M44	Nanopore ionic current / pulse sensing	0	C	C	pore clogging; selectivity
M48	QCM / piezoelectric mass sensor	6	C	C	selective adhesion / drift
Reference / analytical					
M26	FTIR / micro-FTIR / ATR-FTIR	1,705	X	X	specialist instrument; useful external reference
M27	Raman / micro-Raman spectroscopy	664	X	X	specialist instrument; useful external reference
M28	Surface-enhanced Raman (SERS)	61	X	X	nanostructured substrates/laser
M29	Py-GC-MS / thermal decomposition	127	X	X	polymer mass, destructive/lab
M30	SEM / TEM electron microscopy	431	X	X	fixed equipment; morphology
M31	Dynamic light scattering / nanoparticle tracking	29	X	X	nanoparticle proxy, no polymer proof
M32	Plasmonics / SPR / LSPR sensing	5	C	C	substrates/selectivity/expense
M33	Fluorescence tagging / immunoassay	7	X	C	reagents and field stability

Table continued: expanded method identification

ID	Method family	Papers	A	B	Main barrier
M42	NIR / SWIR / near infrared spectroscopy	23	C	C	spectral sampling & water interference
M43	Laser-induced breakdown spectroscopy (LIBS)	7	X	X	pulsed laser / target preparation
M45	AFM atomic force microscopy	20	X	X	lab-scale nanoscale morphology
Indirect / geospatial					
M34	Turbidity / suspended solids	184	S	S	not selective for MP
M35	Conductivity / resistivity / salinity probe	47	S	S	ions dominate; auxiliary covariate
M36	pH / dissolved oxygen / temperature probes	142	S	S	environmental covariates only
M37	RGB / aerial drone imagery	6	S	S	water appearance, not individual MPs
M38	Satellite / remote sensing / spectral indices	33	S	S	surface signal, calibration needed
M39	GIS / geostatistics / spatial interpolation	13	S	S	maps points; not sensing
M40	Machine learning / predictive regression	254	S	S	model does not independently establish polymer

ormalsize **Legend:** **P** = preferred core candidate; **C** = conditional feasibility; **S** = supporting/enabling method, not a stand-alone MP counter; **X** = excluded as routine core under the no-specialist-lab constraint. These are *engineering triage labels*, not published validation status.

Audit trail. The companion `data/Method_to_Paper_Index.csv` identifies every term match by document, DOI and text excerpt. An additional `Unclassified_Technical_Leads.csv` records 137 unmatched candidates with potential instrumentation language, and the larger unmatched file retains the other candidates for reconsideration. These files are not exclusion decisions. Because rule boundaries differ from the earlier screen, the counts of broad categories and detailed categories must never be summed. The candidate pool may also contain methods outside the explicitly defined vocabulary: an unmatched paper is *not* excluded from the literature review.

5.5 Track A: retained candidates versus eliminated methods

Table 3: First-pass decisions for Track A (live boat operation). Priority is conditional on controlled-sample tests.

Method concept	Decision	Reason to retain or use	Why another method is not sufficient
Flow-cell camera + AI [11], [12], [16]	**Core candidate**	Direct particle images, possible events/known observed volume, auditable frames.	Polymer identity unknown; requires de-duplication and good wet optics.
Polarised/laser scatter [19]	**Conditional**	Fast, compact optical signature; little sample handling.	Nonplastic solids, bubbles and changing optical matrix.
Impedance/EIS; resistive pulses [3], [17], [18]	**Conditional**	Compact electronics and continuous-channel possibility.	Electrolyte changes/fouling; electrode chemistry differs from pulse counting.
Holography / flow cytometry [23]	**Conditional**	More features than bulk turbidity; potential high-throughput.	Hardware/optics cost and particle-level truth remain challenging.
Acoustic concentration [24]	**Supporting only**	Upstream enrichment may help low concentrations.	Enrichment does not count or prove MP identity.
Turbidity, conductivity, RGB imagery [8], [14]	**Auxiliary only**	Cheap spatial covariates, interference monitoring, mapping context.	No inherent polymer specificity; must not be called MP particles/L without calibration.
FTIR, Raman, Py-GC-MS [4]	**Eliminate onboard**	Strong offline reference value.	Specialist, slow and maintenance-intensive for an autonomous boat.

5.6 Track B: retained candidates versus eliminated methods

Table 4: First-pass decisions for Track B (known-volume collection; portable shoreline processing).

Method concept	Decision	Reason to retain or use	Limitation to check
Measured-volume filtration + digital microscope / AI [10], [11], [13]	**Core candidate**	Inspect each sample and save images; repeat counts without boat motion.	Time per sample, contamination, particle losses and false plastic IDs.
Static camera + segmentation [12]	**Core analysis stage**	Digitised audit trail and reduced labour.	Representative training set, blind human re-count and operator bias.
Nile Red fluorescence [20], [21]	**Conditional**	Increases contrast for small suspected particles.	Dyes/waste and false positives; not obligatory without safe handling.
EIS, scattering and microfluidics [3], [18], [19]	**Conditional**	Portable bench-side alternative readouts.	May not outperform microscope counts without careful calibration.
Density separation / digestion [10]	**Conditional preparation**	May reduce natural-particle background.	Reagents, time, recovery and safety; avoid unless justified.
FTIR / Raman / Py-GC-MS [4]	**Eliminate routine shoreline**	Occasional external polymer confirmation, if possible.	Requires off-site specialist facility; not a deployable processing stage.

5.7 Transparent weighted comparison *after* the hard gates

The following scores are **illustrative engineering assumptions** on a 1–5 scale, not empirical rankings established by the papers. Six criteria are weighted differently for each workflow: validity of a particles/L output (A/B 30%), portability (A 20%, B 10%), throughput (A 20%, B 10%), independence of specialist labs (A 12%, B 20%), robustness (A 10%, B 15%) and affordability (A 8%, B 15%). A higher number is better. The total is

$$S_{\text{track}} = \sum_{k=1}^{6} w_{k,\text{track}}\, q_k, \quad \sum_{k} w_k = 1, \quad q_k \in \{1,2,3,4,5\}. \tag{1}$$

A score never overrides a failed hard gate. For example, bulk conductivity may be inexpensive and fast, yet remains an auxiliary probe unless a transferably validated model establishes MP concentration. Conversely a micro-FTIR system may provide a strong analytical reference but is excluded from both *routine deployment* tracks.

Table 5: Illustrative track-weighted feasibility of candidate *system concepts*; scores and gates are expert engineering assumptions, NOT study performance. Score range: 1–5.

Concept	A gate	B gate	A / 5	B / 5	Basis / risk
Flow-through camera + AI	P	C	3.62	3.60	Pumped, measured flow; object recognition; count/volume
Portable filtration + microscope / AI	X	P	3.60	3.80	Collect known V; filter; visual count; optional AI
Micro-flow imaging particle analyser	C	C	3.14	3.00	Imaging and morphometry within flow cell
Polarised light scattering	C	C	3.44	3.25	Measure material-related polarisation/scattering contrast
Optical particle scattering / counter	C	C	3.44	3.25	Laser scatter pulses, particle-size proxy
Electrical impedance / EIS	C	C	3.24	3.15	Frequency-dependent electrical response
Resistive-pulse microfluidic	C	C	3.14	3.00	Particle-transit electrical pulses
Nile Red fluorescence + imaging	X	C	2.70	2.75	Controlled portable stain and microscope
Flow cytometry / fluorescence counting	C	C	2.74	2.55	Flow and laser fluorescence/scatter readout
Digital holographic particle imaging	C	C	3.26	3.20	Reconstruct digital holograms and size/categorise
Hyperspectral imaging	C	C	2.34	2.35	Multi-band optical signatures
Acoustic focusing + downstream counting	S	S	2.44	2.50	Enrich suspension; another sensor must count
Ultrasonic direct attenuation	C	C	3.12	2.95	Acoustic losses interpreted as particle-related
Electrochemical functionalised electrode	C	C	2.64	2.60	Electrode binding/redox response
Remote RGB / turbidity proxy	S	S	3.40	3.20	Predict MPs from surface colour/turbidity
Bulk conductivity / resistivity proxy	S	S	3.40	3.20	Predict MPs from ion-dependent water impedance
Micro-FTIR / micro-Raman	X	X	2.50	2.65	Polymer spectrum and particle ID
Py-GC-MS polymer mass	X	X	2.20	2.35	Destructive laboratory polymer mass quantification
SEM / TEM	X	X	1.50	1.60	Electron morphology imaging

5.8 Decisions carried into the prototype phase

The first build should demonstrate **Track B: known-volume collection, controlled shoreline imaging and a GPS-linked provisional concentration map**; its low-infrastructure operation makes it a practical validation baseline. **Track A: flow-through camera/AI** is the main live candidate because it offers an interpretable path from particle image to event count. **EIS or resistive pulse** can be tested in parallel but advances only if blinded MP-vs-silt and salt-sensitivity experiments yield reliable discrimination. Polarised scattering and Nile Red remain options if they improve specificity without unacceptable overhead. This is a *screening decision*, not a finding that any method already satisfies the stakeholder's true-MP validation requirement.

5.9 Camera + AI: counting from collected or flowing water

Lee *et al.* developed skip-connection U-Net architectures for automated particle counting in microscopy images [11]. Xu and Wang examined neural segmentation of urban-water MPs; these studies assess *image segmentation*, not yet continuous MP particles/L accuracy on a boat [12]. Choran and Ormeci investigated micro-flow imaging, including particle shape and size information in flowing samples, with bubble and matrix-related limitations [16]. A complementary flow-cytometry classification study highlights the possibility of discriminating natural particles using additional optical signatures and machine learning, at the cost of more elaborate instrumentation [23]. These studies support the core concept of Track A and the image-analysis step of Track B, but they do not prove either full workflow meets the stakeholder target without further tests.

5.10 Impedance/EIS: possible boat-mounted electrical alternative

An electrode/flow-cell assembly applies an electrical excitation and measures a response; conventional EIS uses the complex frequency-dependent impedance $Z(f) = V(f)/I(f)$. A passing particle can perturb a confined measurement field, while dissolved salts, temperature and fouling can also dominate response. Blevins *et al.* survey emerging approaches including impedance [3]. Kim *et al.* used electrochemical/EIS measurements with specialised

$NiMoO_4$ electrodes to detect PVC model MPs [17]. Warraich *et al.* studied a MXene-coated microfluidic *resistance* sensor; a DC bridge is not the same as multifrequency EIS [18]. The electrical path is a *research option* for Track A, but it offers no easy visual audit trail and requires rigorous salinity/particle challenge tests.

5.11 Polarisation, fluorescence, filtration and sample concentration

Polarised light-scattering responses have been investigated to discriminate model plastic particles from other particles, but turbidity and water matrix remain key issues [19]. Nile Red can highlight candidate MPs for camera analysis; stained organics and dye-handling logistics make it more suited to controlled shoreline post-processing than to a completely continuous, reagent-free boat sensor [20], [21]. Portable filtration followed by digital microscopy is simpler but slow per sample and potentially vulnerable to contamination. Integrated microfluidic sample preparation has also been studied, although channel clogging and recovery remain concerns [22]. Acoustic focusing can enrich particles before a downstream detector, but enrichment alone does not identify the particles or establish the source concentration [24].

Table 6: Methods selected from the candidate literature and where they fit the two workflows. *Proposed fit* is an engineering judgement, not a demonstrated boat-ready product.

Method / citation	Principle and reported setting	Track A live boat	Track B shoreline
Camera + U-Net [11], [12]	Image segmentation / particle counts from microscope images	Good AI basis; needs live flow cell, non-plastic rejection	**Very good**; image filtered/settled sample, audit individual detections
Micro-flow imaging [16]	Optical morphology from a flowing sample	**Good conceptual match**; optics/flow control	Useful but less essential than static microscopy
Nile Red + imaging [20], [21]	Dye fluorescence enhances candidate particle contrast	Poor routine fit: reagent and waste handling	**Possible**; needs blanks, staining protocol, false-positive assessment
Impedance / EIS [3], [17]	Electrical response to particles/interfaces	Possible experimental adjunct; high interference risk	Bench-test candidate, not needed for minimal Track B
Electrical microfluidics [18]	Particle-induced bridge signal in restricted channel	Potentially compact; clogging and salinity issues	Optional comparative sensor test
Polarised scattering [19]	Polarisation response from illuminated particles	Potential optical upgrade after camera baseline	Optional discrimination add-on
Filtration + portable microscopy [10], [13]	Isolate/count imaged suspected particles from known volume	Hard to automate continuously	**Core portable reference**; membrane, blank, operator check
FTIR/Raman [4], [13]	Spectral polymer identification	Not realistic as routine student boat payload	Not assumed available; occasional external confirmation desirable
Acoustic focusing [24]	Particle enrichment using acoustic force	Optional complex front-end	Possible but costly/complex compared with filtration

6 New operational screening constraint: 20 m by 20 m in 120 minutes

6.1 What counts as successful completion?

A successful demonstration requires a geo-referenced two-dimensional concentration *estimate* or appropriately labelled suspected-particle proxy for a $20\,\mathrm{m} \times 20\,\mathrm{m} = 400\,\mathrm{m}^2$ survey region no later than 120 minutes after the start of the operation. The clock includes deployment, surveying or collection, shoreline processing where applicable, quality control, and rendering. Sampling density, polymer specificity, particle-size cut-off, volume denominator and uncertainty are separate acceptance criteria. *Neither a colour plot nor 120-minute operation alone proves valid MP concentration.*

The existing 10,833-paper keyword corpus and 49-category methods screen remain unchanged: the new filter is a subsequent **project-level operational gate**, not an additional scientific-paper exclusion or a revised PRISMA count. Table 7 uses engineering estimates; its labels are conditional until demonstrated on site.

6.2 Quantitative survey-time estimates and the sparse-particle problem

Track A example: five 20-m survey transects with 5-m separation and four 5-m cross-links total 120 m. At an assumed 0.25 m/s, ideal path travel takes 8.0 min. Reserving 60 min for sustained measurement and 25 min for setup, turns, QC and map generation gives 93 **min**. The GPS accuracy must be assessed relative to the 5-m lane separation, and this is *not* complete surface coverage; spatial values between tracks are interpolated. At an actual observed flow of 0.50 L/min for 60 min, 30 L is analysed in total, subject to the sensing chamber's effective interrogated fraction.

Track B example: a four-by-four station grid (16 sites) with 1.0 L collected at an actual 1.5 L/min and an assumed 1.2 min handling overhead plus 3 min portable processing per station, together with 18 min for transit, setup, transfer, mapping and QC, totals approximately 95.9 **min**. These timing assumptions require time-and-motion testing. If microscopy takes 5 min instead of 3 min per station, the mission exceeds 120 min. Collecting 1 L at 16 stations means at least 16 kg of retained water unless samples are processed/emptied during the survey or smaller aliquots are taken; the separate boat design therefore budgets a more limited 8 L stored as an initial payload case.

At a cited Semenyih order-of-magnitude concentration near 3.12 particles/L [2], a 1-L sample yields only 3.12 expected particles if the instrument detects all relevant MPs. For a Poisson count, relative counting standard error is approximately $1/\sqrt{N}$: 57% **for a 1-L sample**. About 3.2 L is needed to expect ten particles (roughly 32% counting error), and 6.4 L for twenty (roughly 22%), *before* non-plastic false positives and detection losses. Increasing sample volume can therefore break the 120-minute target even when GPS coverage is fast. These are arithmetic scenarios, not new Semenyih measurements.

Table 7: Two-hour operational filter applied *after* method-family screening. "Conditional" means a feasible architecture on paper but no demonstrated mission compliance.

Method candidate	Track A	Track B	Time/validity limitation
Flow-cell camera + AI	Conditional	Supporting	Must demonstrate effective observed flow and no repeat counts.
Collected filter + microscope/AI	Not live	Conditional	Sample throughput must be ≤ 3 min/station in illustrative 16-point plan.
Polarised scattering	Conditional	Conditional	Fast potential readout, but plastic-vs-mineral and size range must pass validation.
EIS / impedance cytometry	Conditional	Conditional	Volumetric sensitivity, electrolyte dependence, clogging and calibrated volume.
Nile Red + fluorescence	Not routine live	Conditional	Staining, blanks and waste handling must fit the turnaround.
Turbidity / conductivity / RGB	Auxiliary	Auxiliary	Fast but not a direct MP count; requires independently checked calibration.
FTIR / Raman	Excluded onboard	Excluded routine	Specialist laboratory equipment conflicts with present project assumption.
Acoustic concentration	Supporting	Supporting	Time saving only if paired with proven recovery and downstream count.

6.3 New pass/fail gates (G6–G10)

Table 8: Operational gates added to the existing G0–G5 methodology screen.

G6	End-to-end turnaround	Time the *complete* 400-m^2 acquisition-to-map process; pass if $\leq$ 120 min.
G7	Measured volume / counting statistics	Report analysed litres, per-cell counts, size cut-off, confidence intervals and low-count limitation.
G8	Navigation/geolocation	Demonstrate coverage with independent location-error check and intake-time offset correction.
G9	Energy and flotation	Full wet payload, positive reserve buoyancy, peak electrical protection and $\geq$ 2 h runtime with reserve.
G10	Data integrity and map truth	Retain point observations, timestamps, blanks, uncertainty, and a clear suspected-MP/proxy/validated legend.

6.4 Conclusion after applying the two-hour constraint

Track B is the lower-risk first demonstration if a timed portable workflow achieves no more than three minutes of processing per 1-L sample (or fewer stations/alternative sampling design), while managing stored-water mass. **Track A has the better scaling potential** for denser observations but passes only when the camera/electrical sensor actually interrogates enough water for a meaningful concentration estimate. At Semenyih-like low MP counts, *accuracy and counting volume are more limiting than the boat's travel speed*. The student team should first demonstrate a map of measured or suspected particles with uncertainties; claim polymer-validated MP concentrations only when appropriate reference evidence is available. The accompanying `Boat_Critical_Components/boat_components.tex` translates G8–G9 into a prototype hull, controller, thruster, pump, battery and radio selection. `data/2h_Method_Operational_Gates.csv` records these project-level decisions without modifying the original bibliometric screen.

7 Direct comparison: strengths, weaknesses and technical risks

Table 9: Trade-offs for the two deployment tracks. Advantages are conditional on successful validation.

Criterion	Track A: live flow-cell measurement	Track B: collected sample + shore processing
Main advantage	Near-continuous spatial coverage and immediate feedback; routes can be adapted	Easier to build, clean, inspect, re-count and troubleshoot; lower payload complexity
Main disadvantage	Optical flow stability, power draw, wet electronics, fouling and duplicate-frame counting	Delayed map; sampling is discrete; more hands-on processing
Typical sensors	Camera + light + flowmeter; optional polariser/EIS and water-quality probe	Portable microscope + filtration cassette + camera/AI; optional dye
Volume denominator	Must calibrate *observed* fluid volume, not merely pump nominal throughput	Known collected volume; correct for subsampling and filter losses
Plastic specificity	Difficult in untreated turbid water; classifier may recognise natural fragments as plastic	Better visual access and optional stain, but still not spectroscopic polymer proof
Data integrity	Synchronise intake time, transit delay, frames, GPS and flow	Stable sample ID linked to GPS/time; track handling and filter IDs
Field practicality	More integrated engineering and maintenance risk	More achievable without lab access or expensive instrumentation
Map output	Updated continuously with QC; gaps where detector cannot perform	Heat map after batch processing; retain sampling points and extrapolation uncertainty

For Track A, the denominator must correspond to *fluid actually observed* by the image sensor: $\widehat{C}_A = N_{\text{unique,classified}}/V_{\text{imaged}}$. For Track B, $\widehat{C}_B = (N_{\text{counted}} - N_{\text{blank}})/(V_{\text{collected}}\, r)$ only when a defensible recovery fraction r is measured and the entire analysed fraction is accounted for. Until polymer specificity is validated, both concentrations are *suspected MP-like counts per litre*. Blank correction and recovery uncertainty must not be silently assumed.

7.1 Heat-map generation and communication

The software takes $(\text{lat}, \text{lon}, t, V, N, \text{QC})$ from either track, computes a concentration estimate and displays *actual measured locations* before optional spatial interpolation. Each map needs a legend with units, a declared size cutoff, an explicit timestamp/survey period, sample density and unsupported areas. Interpolate only within supported coverage. Compare prediction values against a subset of *withheld* sample stations; do not validate the interpolation

using exactly the same stations on which it was fitted. Real-time maps and shore-processed maps should use the same schema so their performance can be compared directly.

8 Limitations of this working literature review

The evidence base is a *preliminary title/abstract candidate pool*. Automated categories overlap; a matched keyword does not establish experimental deployment, evidence quality or causal relationships. Selected cited papers illustrate operating principles and performance issues but are not a quantitative meta-analysis of all 10,833 papers. Detailed performance figures and protocols must be checked against full texts before a final thesis. The proposed tracks, tests and prioritisation are engineering recommendations motivated by the stakeholder brief and the reviewed candidate literature.

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